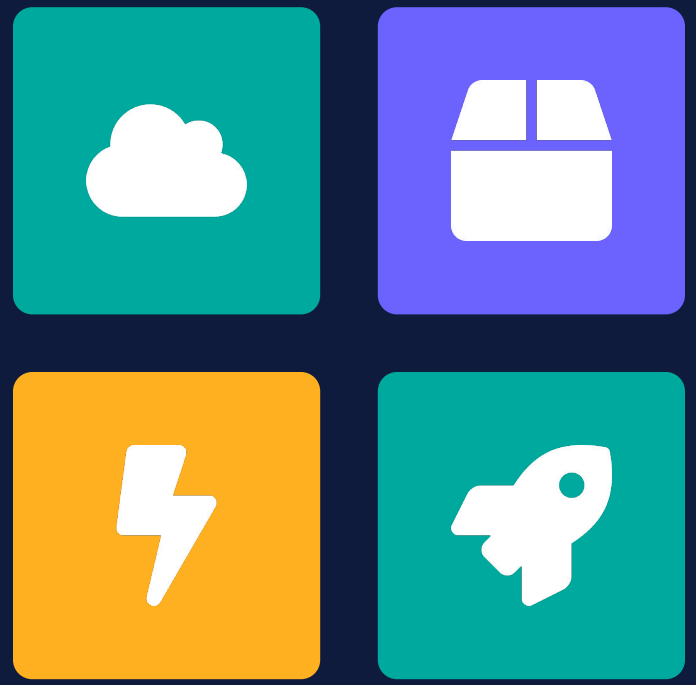
Applications in the Cloud

Going serverless with Cloud Run and Cloud Run functions

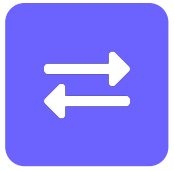
Module 6 · Instructor-led session



By the end of this module, you will be able to



Explain what “serverless” means and who manages what



Compare the container-based and source-based Cloud Run workflows



Describe how Cloud Run billing works and why idle time is free



Decide when to choose Cloud Run functions over Cloud Run

Session plan

Concepts & demo	**35 min**
Knowledge check	**15 min**
Hands-on lab	**30 min**
Q&A / wrap-up	**10 min**

Our journey so far



Virtual machines	Networks	Storage	Containers	Applications
Completed	Completed	Completed	Completed	You are here

So far we provisioned the building blocks ourselves. Now we let Google run them for us, so we can focus purely on code.

Serverless shifts who manages what

	Compute Engine	GKE	Cloud Run
Application code	You	You	You
Container / runtime	You	You	You
Scaling	You	Shared	Google
Operating system	You	Shared	Google
Servers & hardware	Google	Google	Google



With Cloud Run you bring only your code or container; the platform does the rest.

01

Cloud Run

Run any stateless container, fully managed



Cloud Run at a glance

A fully managed platform that runs stateless containers in response to web requests or events.



Stateless containers

Package your app any way you like; each instance keeps no state between requests.



Truly serverless

No clusters, VMs or OS patching. Deploy and get an HTTPS endpoint.



Built on Knative

Open API and runtime on Kubernetes, so workloads stay portable.



Scales to zero, fast

Instances spin up and down almost instantly based on traffic.

Path 1: container-based deployment

1



Write the code

Any language. The app starts a server that listens for HTTP requests on a port.

2



Build the image

Package the app and its dependencies into a container image.

3



Push to Artifact Registry

Store the image so Cloud Run can pull it.

4



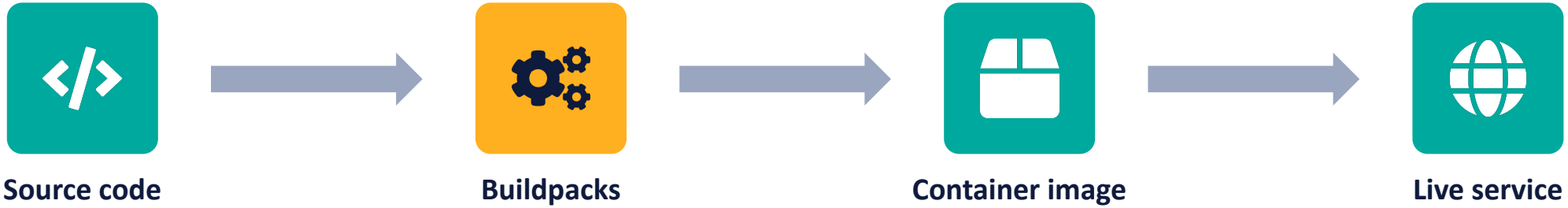
Deploy

Cloud Run starts containers on demand to serve traffic.



Result: a unique HTTPS URL. Cloud Run adds and removes container instances automatically so every incoming request is handled.

Path 2: source-based deployment



Container-based

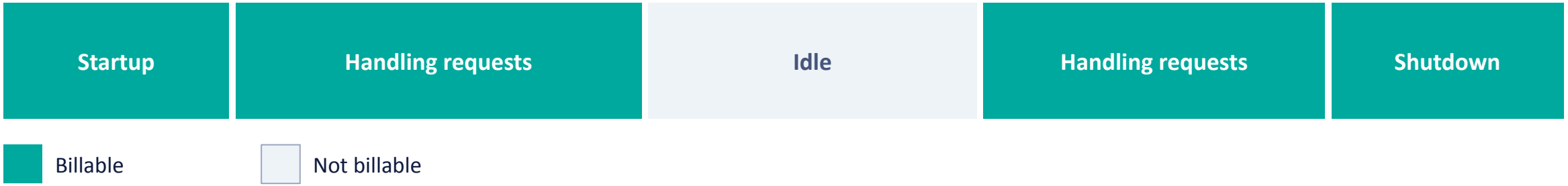
- Full control over the image
- Add custom system packages
- Reuse existing Dockerfiles

Source-based

- Fastest path from code to URL
- Google builds the image for you
- Consistent, secure, well-configured builds

Pay only while your container is working

Lifespan of one container instance (request-based billing)



100 ms

billing granularity for CPU and memory

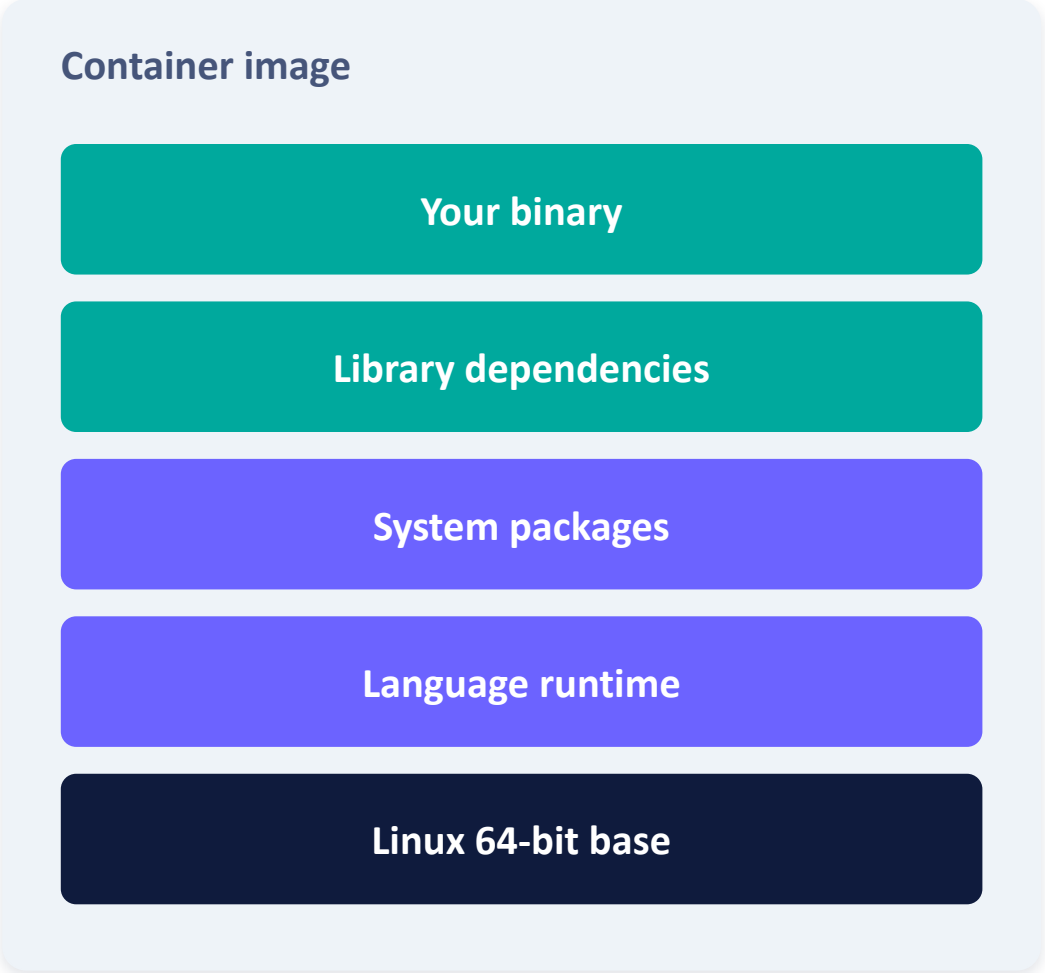
$0

compute cost when no requests are served

+ fee

a small charge per million requests

If it runs on 64-bit Linux, it runs on Cloud Run



Popular languages

Java
Python
Node.js
Go
PHP
C++

Less common, still welcome

COBOL
Haskell
Perl

The only rule: the app must handle web requests.

Where Cloud Run shines



Websites & web apps

Full applications with many routes, templates and user sessions stored externally.



APIs for mobile & web

REST or gRPC back ends that scale with demand.



Microservices

Split a monolith into independently deployed services.



Internal tools & webhooks

Low-traffic admin apps that cost nothing when unused.

02

Cloud Run functions

Small pieces of code that react to events



The problem: event-driven work inside an app



- Convert to a standard format
- Create thumbnails in several sizes
- Store each file in a repository

Built into the main app, this logic needs compute capacity reserved for it, whether uploads happen every millisecond or once a day.

Cloud Run functions: what you get



Event-driven & lightweight

Asynchronous compute that wakes up when something happens.



Single-purpose

Build apps from bite-sized pieces of business logic.



No servers or runtime

Nothing to provision, patch or keep running.



Glue for cloud services

Connect and extend other Google Cloud services.



Billed per 100 ms

You pay only while your code is actually running.



Many languages

Node.js, Python, Go, Java, .NET, Ruby and PHP.

Two ways to trigger a function

Synchronous

The caller waits for the result. Good for webhooks and simple APIs.



HTTP request →  **Function runs** → **Response returned to the caller**

Asynchronous

Nobody waits. Good for processing uploads, messages and data changes.



Cloud Storage or Pub/Sub event →  **Function runs** → **Work completes in the background**

Choosing between the two

	Cloud Run	Cloud Run functions
What you deploy	A container image or source for a whole service	A single function (source code)
Best for	Web apps, APIs, microservices with many endpoints	One task reacting to one event
Triggered by	HTTP requests or events	HTTP requests or Cloud Storage / Pub/Sub events
Language rule	Any binary compiled for 64-bit Linux	Supported runtimes (Node.js, Python, Go, Java, .NET, Ruby, PHP)
Shared traits	Serverless, autoscaling, pay-per-use	Serverless, autoscaling, pay-per-use
Example	A photo-sharing website	Make thumbnails after each upload

Rule of thumb: if it is a whole service, use Cloud Run. If it is one reaction to one event, use a function.

Key takeaways

1. Serverless means Google runs the infrastructure; you focus on code.
2. Deploy to Cloud Run from a container image or straight from source with Buildpacks.
3. Request-based billing charges in 100 ms slices; idle instances cost nothing.
4. Use Cloud Run for whole services and Cloud Run functions for small, event-driven tasks.

03

Knowledge check

7 questions · each followed by an answer slide



Question 1

Choose one

A team needs to host a containerized HTTP API that gets no traffic overnight. They want no servers to manage and no compute cost while idle. Which service fits best?

A Compute Engine VM with an autoscaler

B GKE Standard cluster

C Cloud Run

D Cloud Storage static website

Question 1: answer

A team needs to host a containerized HTTP API that gets no traffic overnight. They want no servers to manage and no compute cost while idle. Which service fits best?

A Compute Engine VM with an autoscaler

B GKE Standard cluster

C Cloud Run

D Cloud Storage static website



Why?

Cloud Run runs stateless containers without any infrastructure to manage and scales to zero, so overnight idle time costs nothing for compute.

Question 2

Choose one

When you deploy source code directly to Cloud Run, which open-source project builds the container image for you?

A Buildpacks

B Knative

C Pub/Sub

D Cloud Logging

Question 2: answer

When you deploy source code directly to Cloud Run, which open-source project builds the container image for you?

A **Buildpacks** ✓

B Knative ✗

C Pub/Sub ✗

D Cloud Logging ✗



Why?

In the source-based workflow, Cloud Run uses Buildpacks to detect the language, install dependencies and package the code into a secure, consistent container image.

Question 3

Choose one

With Cloud Run's request-based billing, when are you charged for container CPU and memory?

A For every hour the service is deployed

B Only when you push a new image version

C For each GB of code stored in the service

D While an instance starts up, handles requests or shuts down

Question 3: answer

With Cloud Run's request-based billing, when are you charged for container CPU and memory?

A For every hour the service is deployed

B Only when you push a new image version

C For each GB of code stored in the service

D While an instance starts up, handles requests or shuts down



Why?

You pay only while an instance is starting, serving requests or shutting down, in 100 ms increments, plus a small per-request fee. Idle time is free.

Question 4

Choose one

Your company has a legacy COBOL program it wants to expose as a web service. What must be true for it to run on Cloud Run?

A It must be rewritten in Python, Java or Go

B It must be compiled for 64-bit Linux and handle web requests

C It must be packaged as a Windows executable

D It must first be converted into a Cloud Run function

Question 4: answer

Your company has a legacy COBOL program it wants to expose as a web service. What must be true for it to run on Cloud Run?

A It must be rewritten in Python, Java or Go

B It must be compiled for 64-bit Linux and handle web requests

C It must be packaged as a Windows executable

D It must first be converted into a Cloud Run function



Why?

Cloud Run runs any binary compiled for 64-bit Linux, including COBOL, Haskell and Perl, as long as the application handles web requests.

Question 5

Select two

Which TWO statements about Cloud Run functions are true?

A Functions can be triggered by Cloud Storage events and Pub/Sub messages

B You must provision a VM for each function

C You are billed only while your function code is running

D Functions can only be written in Java

Question 5: answer

Which TWO statements about Cloud Run functions are true?

A **Functions can be triggered by Cloud Storage events and Pub/Sub messages** ✓

B You must provision a VM for each function ✗

C **You are billed only while your function code is running** ✓

D Functions can only be written in Java ✗



Why?

Cloud Run functions respond to HTTP requests and to events such as Cloud Storage changes or Pub/Sub messages, and you pay in 100 ms increments only while the code runs.

Question 6

Choose one

Which workload is the BEST fit for Cloud Run functions rather than Cloud Run?

A A customer-facing online store with dozens of pages

B A multi-endpoint REST API for a mobile app

C A web app that needs custom system libraries

D Sending a confirmation email each time an order message is published to Pub/Sub

Question 6: answer

Which workload is the BEST fit for Cloud Run functions rather than Cloud Run?

A A customer-facing online store with dozens of pages

B A multi-endpoint REST API for a mobile app

C A web app that needs custom system libraries

D Sending a confirmation email each time an order message is published to Pub/Sub

Why?

One small task reacting to one event is the classic Cloud Run functions pattern. The other three are full services that suit Cloud Run better.

Question 7

Choose one

Which way of invoking a Cloud Run function makes the caller wait for the result (synchronous execution)?

A A message published to a Pub/Sub topic

B A new object uploaded to Cloud Storage

C An HTTP request from a client

D An object deleted from a Cloud Storage bucket

Question 7: answer

Which way of invoking a Cloud Run function makes the caller wait for the result (synchronous execution)?

A A message published to a Pub/Sub topic

B A new object uploaded to Cloud Storage

C An HTTP request from a client

D An object deleted from a Cloud Storage bucket

Why?

HTTP invocation is synchronous: the client sends a request and waits for the response. Cloud Storage and Pub/Sub events trigger functions asynchronously.

Hands-on lab: Hello Cloud Run

Goal: build a simple containerized application image and deploy it to Cloud Run.



Step 1

Create a small web app



Step 2

Build the container image



Step 3

Store it in Artifact Registry



Step 4

Deploy to Cloud Run



Step 5

Open the HTTPS URL



Tip: when you finish, delete the service and image to keep your lab project clean.

Questions & discussion

Thank you for completing Module 6: Applications in the Cloud

